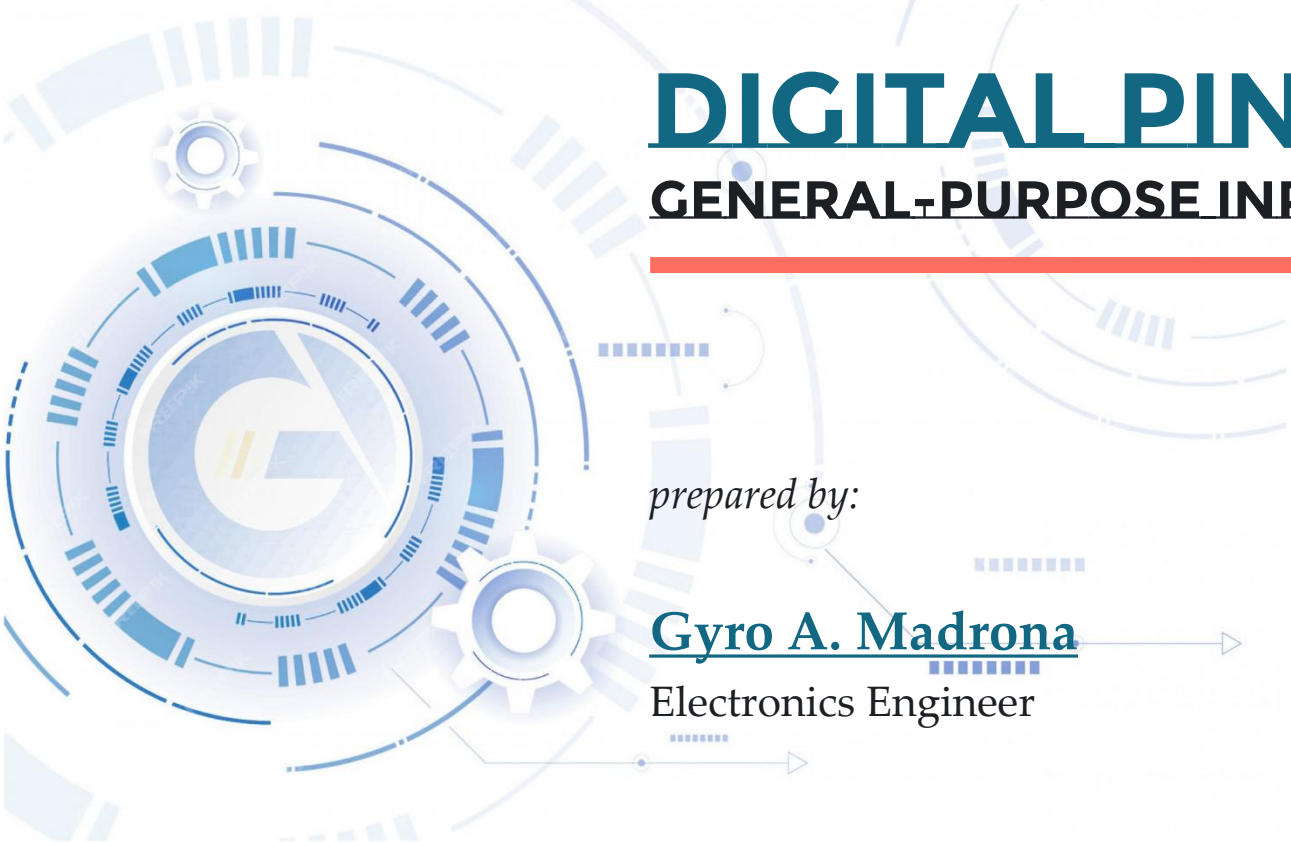




DIGITAL PINS

GENERAL-PURPOSE INPUT/OUTPUT

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Digital Output

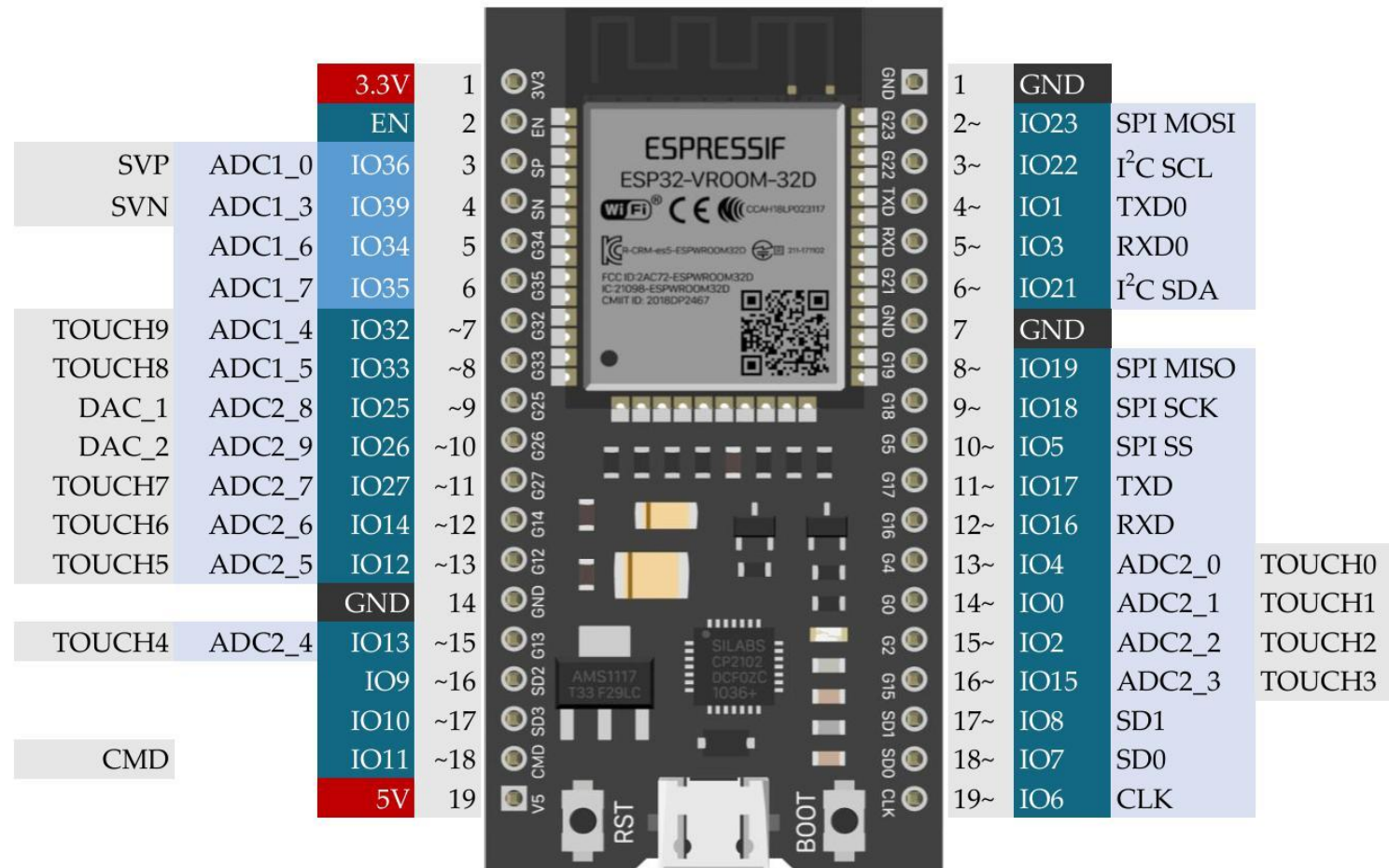
Digital Input



DIGITAL PINS



DEVKITC GPIOs MAP



GPIO voltage: 3.3V (NOT 5V tolerant)

GPIO current: 12-20 mA (40 mA max)

[ESP32-DevKitC-1 - - — Arduino ESP32 latest documentation](#)

BASIC STRUCTURE OF THE ARDUINO FRAMEWORK

Digital_Output.ino

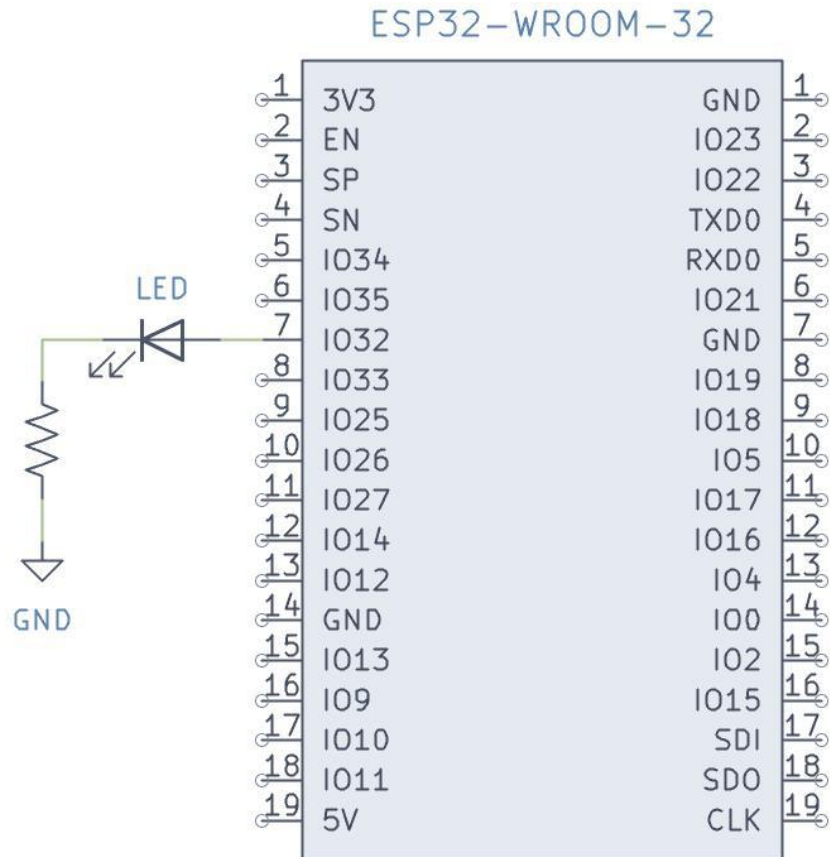
```
1  /*=====
2  | | | | | | | DIGITAL OUTPUT LED
3  =====
4  Description:
5  | This program blinks an LED connected to GPIO32 with
6  | a 1-second interval.
7  |
8  Programmer:
9  | Gyro A. Madrona
10 |
11 Date:
12 | 24 July 2026
13 -----*/
14
15 void setup() {
16     // set GPIO32 as output pin
17     pinMode(32,OUTPUT);
18 }
19
20 void loop() {
21     // LED ON
22     digitalWrite(32,HIGH);
23     delay(1000);           // 1000ms
24
25     // LED OFF
26     digitalWrite(32,LOW);
27     delay(1000);
28 }
```

void setup() – executed **only once** when ESP32 is powered on, reset, or restarted.

void loop() – contains the main logic of the program that the ESP32 performs over and over again.



DIGITAL OUTPUT



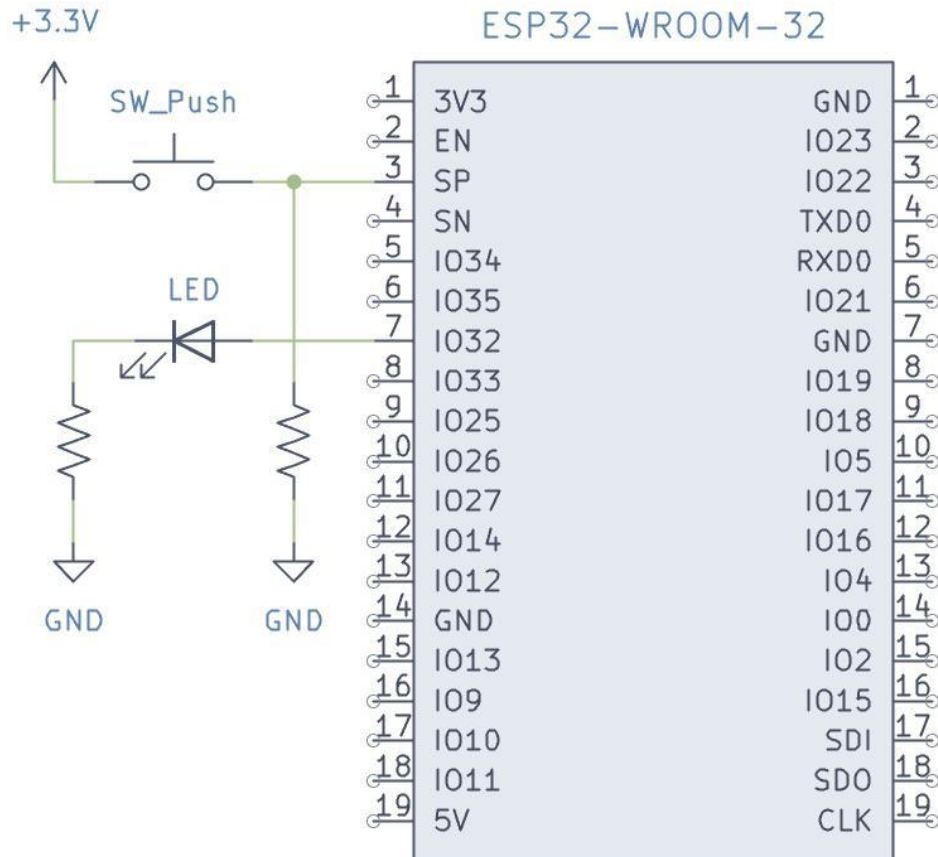
`pinMode()` – configure a GPIO pin as an **INPUT** or **OUTPUT**.

`digitalWrite()` – set a digital output pin to either **HIGH** or **LOW**.

`delay()` – pauses the program for a specified amount of time (in **milliseconds**).



DIGITAL INPUT



`pinMode()` – configure a GPIO pin as an **INPUT, OUTPUT, INPUT_PULLUP, or INPUT_PULLDOWN.**

note: input only pins (GPIO-34,35,36,39) do not have internal pull-up or pull-down resistors.

`digitalRead()` – reads the current logical level of a digital pin and returns either **HIGH** or **LOW**.

LABORATORY

